

The LLM Architecture Decade

10 architectures that defined the last 5 years of language models — and where the field is heading next.

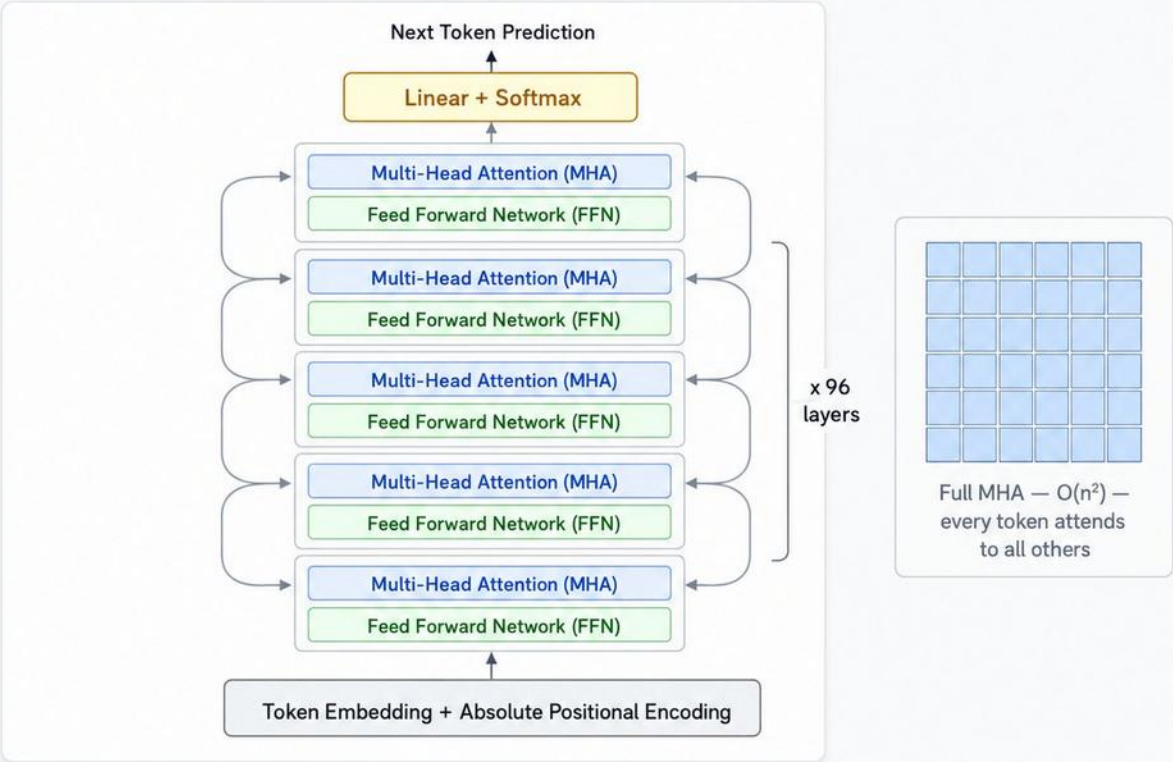


Swipe →

GPT-3

OpenAI · May 2020 · 175B Parameters

SCALE = EMERGENT CAPABILITY



INNOVATION

In-context learning —
steer with prompts, no
fine-tuning needed

PROBLEM SOLVED

One model for all NLP tasks —
no task-specific training

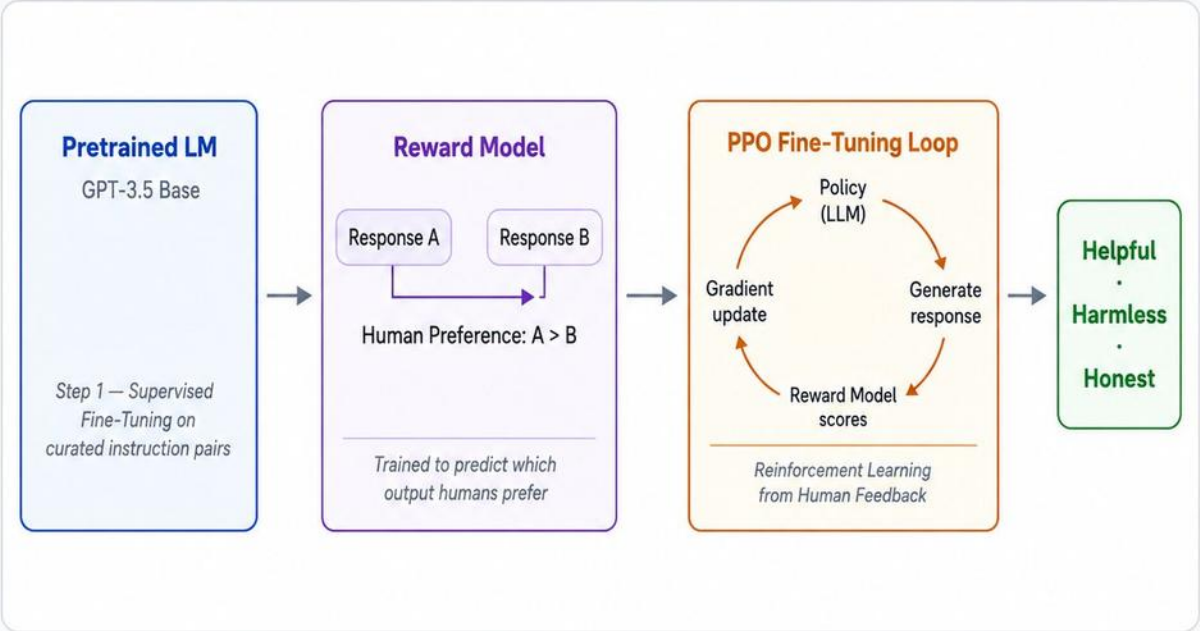
IMPACT

Sparked the LLM race.
First API access to frontier
AI for developers

ChatGPT / InstructGPT

OpenAI · November 2022 · RLHF Fine-Tuning

ALIGNMENT — MAKING LLMs STEERABLE



INNOVATION

RLHF — humans rank outputs to define what good behavior means

PROBLEM SOLVED

Powerful but unpredictable models made safe for public use

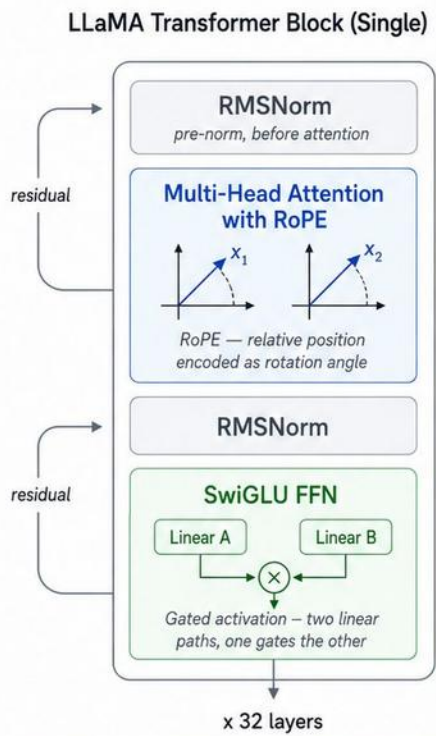
IMPACT

100M users in 2 months. Redefined what a software product could be

LLaMA 1

Meta · February 2023 · 7B–65B Parameters · Open Weights

DEMOCRATIZATION OF FRONTIER AI



Architecture Comparison

Component	GPT-3	LLaMA 1
Normalization	LayerNorm	RMSNorm
Position Encoding	Absolute	RoPE
Activation	GELU	SwiGLU
Norm position	Post-norm	Pre-norm

This exact stack became the default open-model recipe

INNOVATION

RMSNorm + RoPE + SwiGLU – a recipe the entire field adopted

PROBLEM SOLVED

Frontier models locked behind APIs — now anyone could run, study, fine-tune

IMPACT

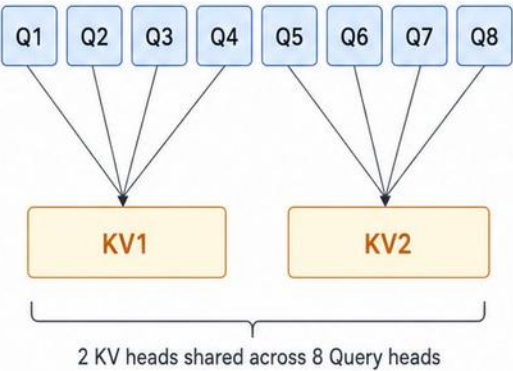
Alpaca, Vicuna, WizardLM emerged within weeks. Open-source LLM ecosystem born

Mistral 7B

Mistral AI · September 2023 · 7B Parameters · Apache 2.0

EFFICIENCY-FIRST ARCHITECTURE

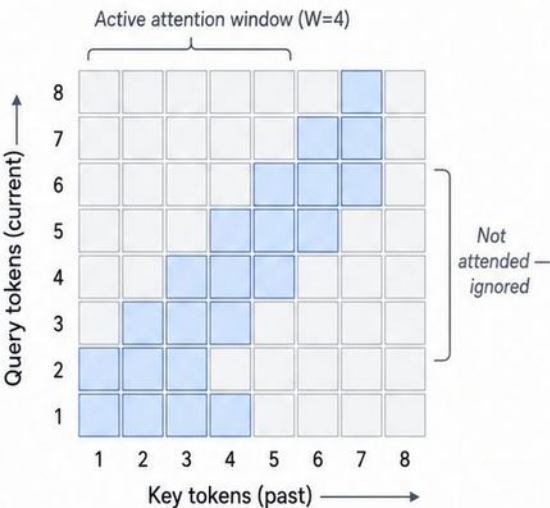
Grouped Query Attention (GQA)



KV cache reduced 4x vs full MHA

Inference memory drops dramatically — critical for serving

Sliding Window Attention (SWA)



$O(n^2) \rightarrow O(n \cdot W)$ — linear scaling

Context built up gradually through layer depth

INNOVATION

GQA reduces KV cache.
SWA reduces attention cost.
Both now in every modern LLM

PROBLEM SOLVED

7B models were slow and memory-heavy to serve.
Mistral 7B outperformed LLaMA 2 13B

IMPACT

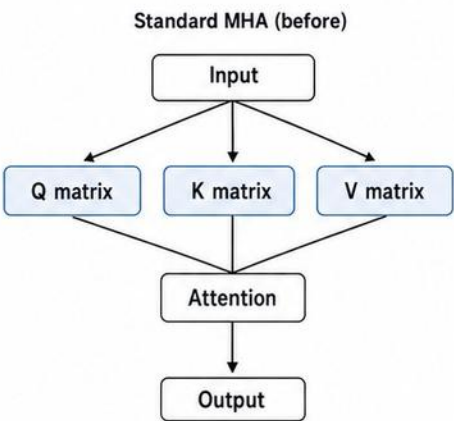
Proved architecture choices matter as much as scale.
First fully commercial open model

DeepSeek V2

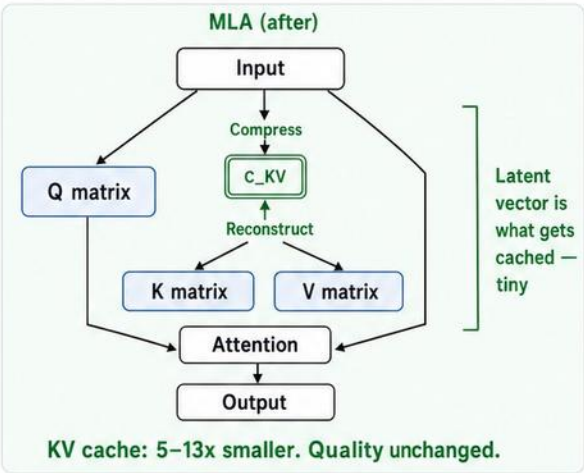
DeepSeek AI · May 2024 · 236B Total / 21B Active Parameters

MLA — THE KV CACHE REVOLUTION

Multi-head Latent Attention (MLA)



KV cache: 1 entry per head per layer — grows large



KV cache: 5–13x smaller. Quality unchanged.

Mixture of Experts (MoE)



236B total parameters. Compute cost equivalent to a 21B dense model.

INNOVATION

MLA compresses K and V into a low-rank latent — reconstructed at inference time

PROBLEM SOLVED

At scale, KV cache memory was the dominant bottleneck, not compute

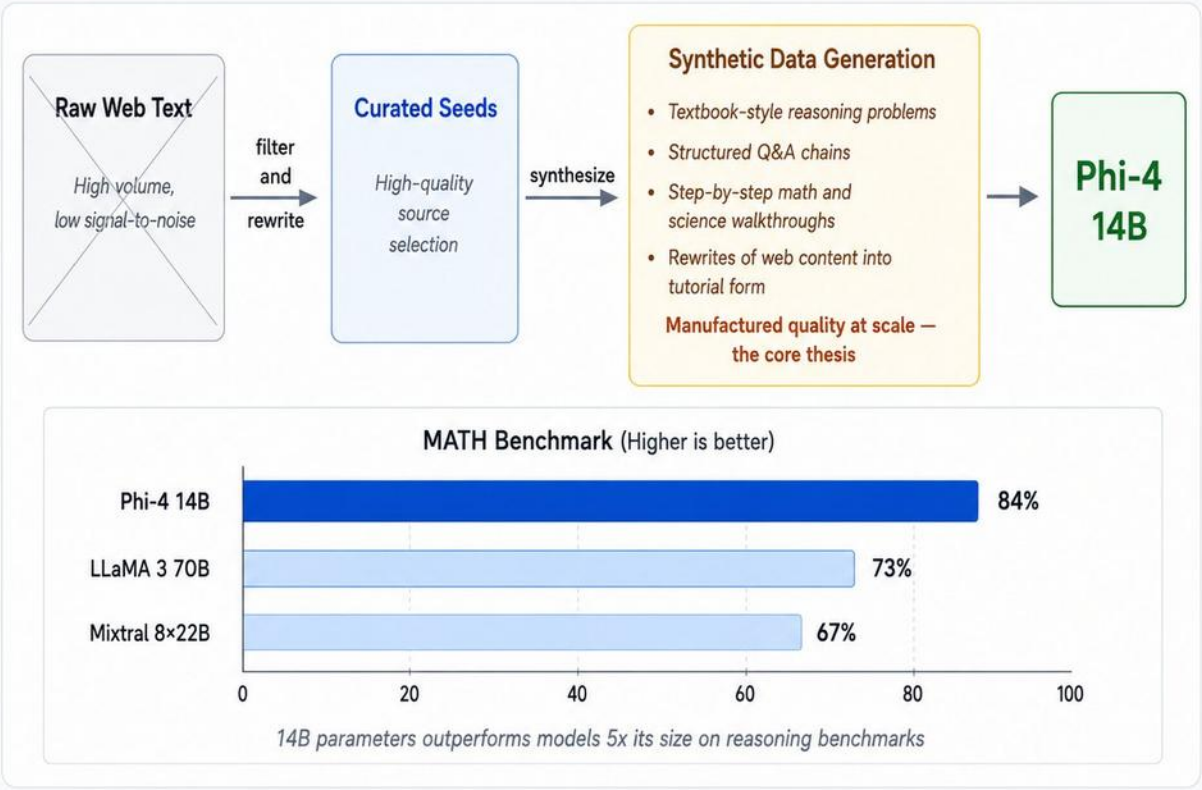
IMPACT

MLA adopted by DeepSeek V3, R1, Kimi K2, and Mistral Large 3 within one year

Phi-4

Microsoft · December 2024 · 14B Parameters · MIT License

DATA QUALITY AS A SCALING AXIS



INNOVATION	PROBLEM SOLVED	IMPACT
Synthetic data pipeline — curate and manufacture training quality, not just filter for it	The assumption that scale always beats quality — overturned at 14B	Data curation became a research discipline. Training pipeline is now a competitive moat

DeepSeek V3

DeepSeek AI · December 2024 · 671B Total / 37B Active Parameters

FRONTIER QUALITY AT FRACTION OF COMPUTE COST

A. Dense Prefix (3 layers)

RMSNorm

Full MHA

Stable gradient warm-up before MoE routing begins

B. MoE Layers (x58)

RMSNorm

MLA
Multi-head Latent Attention

RMSNorm

Shared Expert
always active

Routed Experts

1 2 3 4 5 6 ... 256

top-2 of 256

residual

residual

C. MTP-1 Head (One Additional Prediction)

Multi-Token Prediction — stronger training signal, faster generation

Token t

Token t+1

(main output)

(MTP head)

Training Cost Comparison

Model	Approx. Cost
DeepSeek V3	~\$6M
GPT-4 class	~\$100M+
Llama 3 405B	~\$30M+

Matched GPT-4o performance at 10x lower cost

Nvidia stock fell 17% the day efficiency figures circulated publicly

INNOVATION	PROBLEM SOLVED	IMPACT
MLA + MoE + shared expert + MTP — every component tuned for cost efficiency	Belief that frontier performance requires frontier hardware budgets	Showned algorithmic efficiency can substitute for hardware. Changed industry economics

The LLM Architecture Decade

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DeepSeek R1

DeepSeek AI · January 2025 · 671B Total / 37B Active · MIT License

OPEN REASONING VIA PURE REINFORCEMENT LEARNING

PHASE 1

DeepSeek V3 Base

Pre-trained language model —
predicts next tokens, no reasoning ability

↓ No supervised fine-tuning. No human-annotated reasoning traces.

PHASE 2

Pure Reinforcement Learning on Verifiable Tasks

Correct answer

+reward

Wrong answer

no reward

No human labels. No reasoning templates.

Rule-based — simple, scalable, zero annotation cost

Before (direct)

Q: "What is 847 x 63?"

A: "53,361" —
direct, no process
shown



After (emerged reasoning)

Q: "What is 847 x 63?"

A: "Let me work through this
step by step.
800 x 63 = 50,400
47 x 63 = 2,961
Total = 53,361"

Chain-of-thought emerged from RL — not explicitly trained



PHASE 3

DeepSeek R1 — Open, MIT License

Distilled to 1.5B, 7B, 14B parameter variants —
small models that reason above their size class

INNOVATION

Pure RL on verifiable outcomes —
model learns to think without
being shown how

PROBLEM SOLVED

Reasoning was a closed capability
(OpenAI o1). R1 made it open
and reproducible

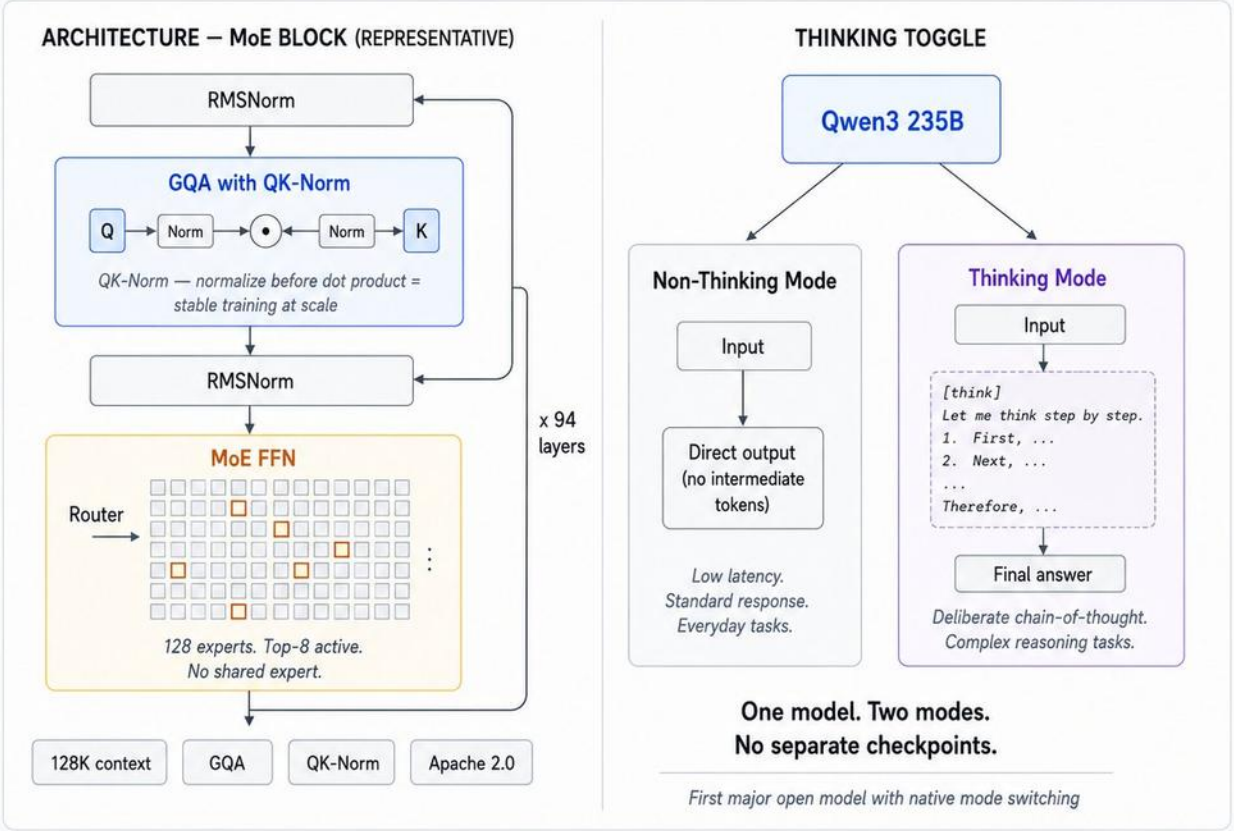
IMPACT

Community distilled R1 into
sub-14B models exceeding much
larger non-reasoning baselines

Qwen3

Alibaba / Qwen Team · April 2025 · 235B Total / 22B Active · Apache 2.0

GLOBAL FRONTIER — THINKING TOGGLE



INNOVATION	PROBLEM SOLVED	IMPACT
QK-Norm for stable MoE training. Thinking toggle without separate model weights	Frontier open AI concentrated in US labs — Qwen3 demonstrated otherwise	GPT-4o class performance. Apache 2.0 commercial use. Innovation is now globally distributed

SSM and Hybrid Attention

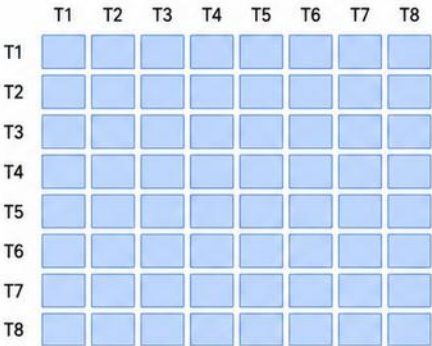
xLSTM · Mamba · DeltaNet · Lightning Attention · 2025 onwards

BEYOND QUADRATIC ATTENTION

Transformer Self-Attention

(every model 2017–2024)

Every token attends to every other token

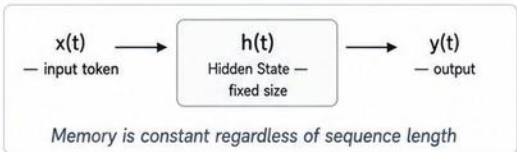


Complexity: $O(n^2)$
2x context length = 4x compute and memory

KV cache: grows linearly with context —
memory bottleneck at 100K+ tokens

Hybrid Architecture

(2025 frontier)



Complexity: $O(n)$
KV cache: near-zero — 68 KiB → 7.9 KiB per token

xLSTM 7B —
pure recurrent, no attention

Nemotron Nano —
Mamba-2 + MoE

Kimi Linear —
DeltaNet + MLA, 1M context

Qwen3.5 —
hybrid as mainline architecture

INNOVATION

Replace most attention layers
with SSMs — linear time,
constant memory regardless
of context length

PROBLEM SOLVED

Quadratic attention made
serving at 100K–1M token
contexts economically
impractical

IMPACT

The active architectural frontier.
Where the next generation of
efficiency gains is coming from

THE PATTERN

Five years. Three shifts.

01 · SCALE

GPT-3 proved that emergent ability comes from parameters and tokens — not task-specific tricks.

02 · ALIGNMENT

RLHF turned raw capability into products. ChatGPT redefined what software feels like.

03 · EFFICIENCY

MLA, MoE, GQA, SSMs — the frontier is no longer just bigger. It's smarter per FLOP, per byte, per dollar.

Which shift surprised you most?

Drop a comment — I'd love to hear which architecture you think is most underrated.